

This Question Paper consists of 39 questions and 9 printed pages.

Roll No.

2	1	0	2	7	3	2	5	3	0	1	5
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Code No. 71/SS/330/D1

COMPUTER SCIENCE
(330)

Set **D1**

Day and Date of Examination

Thursday, 30/04/2026

Signature of Invigilators

1.

2.

General Instructions :

1. Candidate must write his/her Roll Number on the first page of the Question Paper.
2. Please check the Question Paper to verify that the total pages and total number of questions contained in the Question Paper are the same as those printed on the top of the first page. Also check to see that the questions are in sequential order.
3. For the objective type of questions, you have to choose any one of the four alternatives given in the question, i.e., (A), (B), (C) or (D) and indicate your correct answer in the Answer-Book given to you.
4. Making any identification mark in the Answer-Book or writing Roll Number anywhere other than the specified places will lead to disqualification of the candidate.
5. Answer for all questions, including Matching, True or False, Fill in the blanks, etc., are to be given in the Answer-Book only.
6. Write your Question Paper Code No. 71/SS/330/D1, Set D1 on the Answer-Book.



COMPUTER SCIENCE

(330)

Time : 3 Hours]

[Maximum Marks : 60

Note : (i) This Question Paper consists of 39 questions in all.

(ii) All questions are compulsory.

(iii) Marks are given against each question.

(iv) **Section-A** consists of

(a) Question Nos. **1** to **12**—Multiple Choice-type Questions (MCQs) carrying 1 mark each. Select and write the most appropriate option out of the four options given in each of these questions.

(b) Question Nos. **13** to **24**—carry 1 mark each and Questions Nos. **25** to **27** Objective type questions of 2 marks each. Attempt these questions as per the instructions given for each of the questions **13** to **27**.

(v) **Section-B** consists of

(a) Question Nos. **28** to **35**—Very Short Answer-type questions carrying 2 marks each to be answered in the range of 30 to 50 words.

(b) Question Nos. **36** and **37**—Short Answer-type questions carrying 3 marks each to be answered in the range of 50 to 80 words.

(c) Question Nos. **38** and **39**—Long Answer-type questions carrying 4 marks each to be answered in the range of 80 to 120 words.

An internal choice has been provided in some of these questions in **Section-B**. You have to attempt only one of the given choices in such questions.

- (1) Answers of all questions are to be given in the Answer-Book given to you.
- (2) 15 minutes time has been allotted to read this Question Paper. The Question Paper will be distributed at 2:15 p.m. From 2:15 p.m. to 2:30 p.m., the students will read the Question Paper only and will not write any answer on the Answer-Book during this period.

SECTION-A

Note : Question Numbers **1** to **12** are MCQs of 1 mark each.

1. Which attribute of the tag is used to define the source URL of the image displayed on a webpage? 1

(A) source

(B) url

(C) src

(D) path

2. Which among the following allows you to save and close a presentation in PowerPoint? 1

(A) Open a presentation

(B) Close a presentation

(C) Que a presentation

(D) Print a presentation



3. Which HTML tag is used to define an unordered (bulleted) list? 1
 (A) <bl> (B)
 (C) (D) <dl>
4. To compute e^x in C++, which type of header file must be included? 1
 (A) String (B) Mathematical
 (C) Character (D) Text
5. Which of the following is used for displaying the text "WELL DONE" in BOLD in HTML? 1
 (A) <bold> WELL DONE </bl> (B) <bl> WELL DONE </underline>
 (C) WELL DONE (D) <bb> WELL DONE </bb>
6. What is the correct way to close a line break tag in HTML? 1
 (A) </br> (B) <br//>
 (C) No closing tag is required (D) <*br>
7. Find the output produced by the following program segment. Assume all header files are included. 1
- ```
void main()
{
 int x = 1011;
 int*y = &x;
 cout<<*y;
}
```
- (A) 1011 (B) \*1011  
 (C) &1011 (D) &X
8. Which HTML attribute is used to define a border around an image? 1  
 (A) line (B) box  
 (C) border (D) black line
9. Which tag in HTML forces the content to move to the next line? 1  
 (A) <break> (B) <br>  
 (C) <line> (D) <linebreak>
10. Which type of content can be inserted into a LibreOffice Impress slide? 1  
 (A) Text (B) Picture  
 (C) Audio (D) All of the above



11. In word processors like MS Word or LibreOffice Writer, which of the following is supported? 1  
 (A) Creating data (B) Formatting data  
 (C) Printing data (D) All of the above
12. Identify the software that is categorized as spreadsheet software. 1  
 (A) MS Word (B) Impress  
 (C) Lotus 1-2-3 (D) All of the above
13. Name a communication protocol used in networking. 1
14. Fill in the blank : 1  
 The octal equivalent of  $(1001)_{10}$  is \_\_\_\_.
15. What will be the output produced by the below given program if the value of the variable x is 18? 1  

```
#include<iostream.h>
int main()
{
 int x;
 cin>>x;
 y=3
 if (x/y<12)
 cout<<"Morning looks Good";
 else
 cout<<"Evening looks Dull"; return 0;
}
```
16. Fill in the blank : 1  
 The decimal equivalent of  $(ABC)_{16}$  is \_\_\_\_.
17. Which of the heading tags from <h1> to <h6> shows the largest font by default? 1
18. To use  $\log_{10}(x)$  in a C++ program, which library should be included? 1
19. What is the ASCII value of 'B'? 1
20. What does ARPANET stand for? 1
21. Consider the below given program. What output will be produced if the value for the variable ch is entered as 'A'? 1  

```
#include<iostream.h>
void main()
{
 char ch;
 cin>>ch;
 switch(ch)
```



```

{
 case 'A':
 cout<<"A Grade";
 break;
 case 'B':
 cout<<"B Grade";
 break;
 case 'C':
 cout<<"C Grade";
 break;
 case 'D':
 cout<<"D Grade";
 break;
 default:
 cout<<"NO Grade";
}
}

```

22. What will be the output produced by the below given code?

1

```

#include<iostream.h>
void main()
{
 int i=0, s22=0;
 while (i < 4)
 {
 s22=s22+i;
 i++;
 }
 cout<<s22;
}

```

23. What will be the output produced by the below given code?

1

```

#include<iostream.h>
void main()
{
 struct emp
 {
 int eno;
 float sal;
 };
 struct emp E;
 E.eno=9001;
 E.sal=90001.50;
 cout<<E.eno<<E.sal;
}

```





24. What is Statement1 in the below given program? Assume all header files are included.

```
#include<iostream>

void change(int); //Statement1

void main()
{
 int a = 5;
 cout<<"a="<<a<<endl;
 change(a);
 cout<<"a="<<a;

}

void change(int b)
{
 b=10;
}
```

25. Fill in the blanks :

- (a) Writing data on the data file requires \_\_\_\_\_ header file.  
(b) ios :: app opens the file in \_\_\_\_\_ mode.

26. State whether the following statements are True or False :

- (a) A pointer is a variable that refers to the location of another variable.  
(b) Object pointers are used to create or refer objects during runtime.

27. Fill in the blanks :

- (a) The smallest unit of a worksheet is known as \_\_\_\_\_.  
(b) \_\_\_\_\_ page of a spreadsheet is known as worksheet.

### SECTION-B

28. What is simplex communication in networking?

**Or**

What is meant by the term 'bandwidth' in data communication?

29. What is an attribute in the context of a database?

**Or**

What is a domain in the context of a database?



30. Write SQL queries for the following based on the below given BUS table :

2

| BUS   |          |      |        |
|-------|----------|------|--------|
| BUSNO | BUSNAME  | YEAR | TYPE   |
| B001  | NEERU    | 2024 | VOLVO  |
| B002  | RAVANA   | 2022 | SCANIA |
| B003  | MAHAVEER | 2019 | VOLVO  |
| B004  | ROLI     | 2024 | TATA   |
| B005  | VIKRAM   | 2019 | BENZ   |

(a) Insert the record B006, LAXMI, 2023, VOLVO.

(b) Display the details of VOLVO buses.

31. Name any *two* components of Writer window.

2

32. Why should an HTML document be saved with the extension.html? Name one browser that helps us to open an HTML document.

2

33. Consider the code fragment given below :

```
#include<iostream.h>
#include<string.h>
struct ITEM
{
char iname[20];
int price;
};

void main()
{
____ // Statement 1
____ // Statement 2

cout<<s1.iname;
cout<<s2.price;

}
```

Complete the above code for the following:

2

Statement 1 : Create two objects s1 and s2 of ITEM.

Statement 2 : Input the value for the variable iname of the structure variable s1.



34. Write SQL queries to create the below given BUS table :

| BUS     |             |             |
|---------|-------------|-------------|
| BUSNO   | VARCHAR(6)  | PRIMARY KEY |
| BUSNAME | VARCHAR(15) |             |
| YEAR    | INT(4)      |             |
| TYPE    | CHAR        |             |

Or

Differentiate between fixed length and variable length data type.

35. Define the following terms (any two) :

1×2=2

- (a) Member Functions
- (b) Object
- (c) :: operator
- (d) Inline Member Function

36. Create a webpage using the following instructions :

3

- (a) Heading 'Examination' on the center of the webpage with the smallest heading font
- (b) The Paragraph "I feel My Exam has gone good" on towards the left of the webpage
- (c) An image to be set having the file name "NIOS2024.jpg" with a height of 6 and width of 3

37. Define the following terms (any three) :

1×3=3

- (a) Primary key
- (b) Physical data model
- (c) Domain
- (d) Degree
- (e) Default constraint

38. (a) Answer the questions from (i) to (iv) based on the below given code (assume all header files are included) :

4

```
class Graduate {
 int Rollno;
 char GName[20];
 float Marks1;
protected:
 void Result();
public:
```





```

Graduate();
void Enroll();
void Display();
};
class Lecturer {
 long TCode;
 char TName[20];
protected:
 float Salary;
public:
 Lecturer();
 void Enter();
 void Show();
};

class Learn: public Graduate, private Lecturer {
 long CCode[10];
 char LearnName[50];
 char StartDate[8], EndDate[8];
public:
 Learn();
 void Commence();
 void CDetail();
};

```

- (i) Write the names of member functions, which are accessible from objects of class Learn.
- (ii) Write the names of all the data members, which are accessible through member function Commence of class Learn.
- (iii) Write the names of all the members, which are accessible from objects of class Lecturer.
- (iv) Which type of Inheritance is illustrated in the above C++ code?

**Or**

- (b) Define Virtual Base Class. Display the structure by taking an example or with syntax.

- 39. (a)** Write a program to create an array of  $N$  elements, accept a number and location from the user, delete the accepted number from the array and print the new array.

4

**Or**

- (b) Write a C++ program that accepts two numbers (a, b) from the user and passes to a function, the function will perform  $(a*a) + (b*b)$  to the main program which will display the output to the user.

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